From Hohenberg–Kohn to a working Kohn–Sham calculation

Detailed proofs, functional derivatives, and a verified SCF example

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Quantum mechanics for many-particle systems
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Continuation of the one- and two-body density-matrix derivations.
The color scheme and teaching boxes follow the supplied weekly templates.

The route from a theorem to an algorithm

1. **Hohenberg–Kohn I:** why a ground-state density fixes a scalar external potential, apart from a constant.
2. **Hohenberg–Kohn II:** why minimizing an exact density functional recovers the ground-state energy.
3. **Constrained search:** define the functional on a usable density domain and handle degenerate states.
4. **Kohn–Sham:** introduce auxiliary orbitals and derive their equations with all constraints and functional derivatives shown.
5. **Implementation:** discretize, build potentials, solve, occupy, mix, test convergence, and evaluate a consistent energy.

Key result

Exact reformulation, approximate functional, finite representation, and incomplete iteration are four different issues.

Where the density-matrix supplement ended

With $x = (\mathbf{r}, \sigma)$ and the ordered-pair convention,

$$\gamma(x; x') = \langle \hat{\psi}^\dagger(x')\hat{\psi}(x)\rangle, \quad P(x, y) = \langle \hat{\psi}^\dagger(x)\hat{\psi}^\dagger(y)\hat{\psi}(y)\hat{\psi}(x)\rangle,$$

$$\mathrm{Tr}\,\gamma = N, \qquad \int \mathrm{d}x\,\mathrm{d}y\, P = N(N-1), \qquad n(\mathbf{r}) = \sum_\sigma \gamma(\mathbf{r}\sigma; \mathbf{r}\sigma).$$

For a local spin-independent interaction,

$$E = T[\gamma] + \int v_{\mathrm{ext}}\, n\, \mathrm{d}^3 r + \frac{1}{2}\int w(\mathbf{r}, \mathbf{r}')P(x, y)\,\mathrm{d}x\,\mathrm{d}y.$$

Many-body connection

For arbitrary states, the local density does not specify the RDMs. DFT obtains a density functional by minimizing over the missing many-body information, rather than discarding its energetic consequences.

Specify the family of Hamiltonians before stating the theorem

Use Hartree atomic units: $\hbar = m_e = e^2/(4\pi\epsilon_0) = 1$.

$$\hat{H}[v] = \hat{T} + \hat{W} + \hat{V}[v], \quad \hat{T} = -\frac{1}{2}\sum_i \nabla_i^2, \quad \hat{W} = \sum_{i<j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}, \quad \hat{V}[v] = \sum_i v(\mathbf{r}_i).$$

- Fix N, particle statistics, $\hat{T}$, $\hat{W}$, the Hilbert space, and boundary conditions. Change only a scalar local potential v.
- Assume a lower-bounded Hamiltonian with an attained ground-state energy. Initially assume a nondegenerate ground state.
- For potential uniqueness, use a continuum potential class with the needed unique-continuation property; molecular Coulomb potentials are included.

A trap to avoid

The ordinary density-only theorem is not a theorem for arbitrary nonlocal external operators or magnetic vector potentials. "Universal" means independent of v within this fixed problem family.

The crucial identity: the external energy uses only the density

For normalized antisymmetric Ψ, with $X = (x_2, \ldots, x_N)$,

$$n_\Psi(\mathbf{r}) = N \sum_{\sigma_1} \int \mathrm{d}X \, |\Psi(\mathbf{r}\sigma_1, X)|^2, \qquad \int n_\Psi \, \mathrm{d}^3 r = N.$$

Permutation symmetry of $|\Psi|^2$ makes all N contributions equal:

$$\begin{aligned} \langle \hat{V}[v] \rangle_\Psi &= \sum_i \int \mathrm{d}x_1 \cdots \mathrm{d}x_N \, v(\mathbf{r}_i) |\Psi(x_1, \ldots, x_N)|^2 \\ &= N \sum_{\sigma_1} \int \mathrm{d}^3 r_1 \, \mathrm{d}X \, v(\mathbf{r}_1) |\Psi(\mathbf{r}_1\sigma_1, X)|^2 \\ &= \boxed{\int v(\mathbf{r}) n_\Psi(\mathbf{r}) \, \mathrm{d}^3 r}. \end{aligned}$$

Only this identity and the variational principle enter the main contradiction.

Rayleigh–Ritz: exactly when is the inequality strict?

For a normalized trial state, $E_0 \leq \langle \hat{H} \rangle_\Phi$. For a discrete spectral expansion $|\Phi\rangle = \sum_k c_k |\Psi_k\rangle$,

$$\langle \hat{H} \rangle_\Phi - E_0 = \sum_k |c_k|^2 (E_k - E_0) \geq 0.$$

- Equality holds precisely when the trial state lies in the ground-state eigenspace. The same conclusion holds with continuous spectrum included.
- With a nondegenerate ground state, equality means $\Phi = e^{i\theta}\Psi_0$.
- A different normalized *ray*, not merely a different phase, gives a strict inequality.

Key result

To write $E_0 < \langle \hat{H} \rangle_\Phi$, we must know that Φ is not a ground state of this Hamiltonian. Non-degeneracy alone does not establish that two different potentials have different ground states.

Why the external potential is determined only up to a constant

If $v'(\mathbf{r}) = v(\mathbf{r}) + C$, then

$$\hat{H}[v'] = \hat{H}[v] + NC, \qquad E_k[v'] = E_k[v] + NC.$$

Every eigenstate and every density is unchanged.

A gauge choice

For an isolated atom or molecule one often fixes the zero by specifying the potential at infinity. Other boundary conditions require their own convention.

n_0 can determine v only modulo an additive constant.

Changing the constant also shifts Kohn–Sham eigenvalues and the number multiplier μ. Density and energy differences at the same particle number remain invariant.

The common-state lemma closes a gap in the short proof

Suppose the same nonzero Ψ is an eigenstate of $\hat{H}[v]$ and $\hat{H}[u]$. Subtract the two Schrödinger equations:

$$\left[\sum_i (v-u)(\mathbf{r}_i)\right]\Psi = (E_v - E_u)\Psi.$$

For the stated continuum potential class, a relevant nonzero spatial component of Ψ is nonzero almost everywhere. Hence

$$\sum_i \Delta v(\mathbf{r}_i) = E_v - E_u \quad \text{almost everywhere.}$$

Fix the other coordinates and vary $\mathbf{r}_1$:

$$\Delta v(\mathbf{r}_1) = E_v - E_u - \sum_{i=2}^{N} \Delta v(\mathbf{r}_i) = C, \qquad NC = E_v - E_u.$$

A trap to avoid

Nodes on sets of measure zero are allowed. Infinite barriers, disconnected configuration spaces, lattice models, or arbitrary singular potentials need separate analysis. This is a stated assumption, not a proof of unique continuation.

HK I: assume two inequivalent potentials give the same density

Let v and u differ by more than a constant. Suppose

$$\hat{H}[v]\Psi_v = E_v\Psi_v, \qquad \hat{H}[u]\Psi_u = E_u\Psi_u, \qquad n_{\Psi_v} = n_{\Psi_u} = n_0.$$

Both ground states are normalized and nondegenerate.

1. The common-state lemma implies that their rays are different.
2. Therefore Ψ_u is a strict variational trial state for $\hat{H}[v]$, and vice versa.
3. The Hamiltonians differ only by the one-body operator

$$\hat{H}[v] - \hat{H}[u] = \sum_i [v(\mathbf{r}_i) - u(\mathbf{r}_i)].$$

We can now justify every line of the usual short proof.

HK I: derive the first inequality line by line

Use Ψ_u as a trial state for the v problem:

$$
\begin{aligned}
E_v &< \langle \Psi_u | \hat{H}[v] | \Psi_u \rangle && \text{strict Rayleigh–Ritz} \\
&= \langle \Psi_u | \hat{H}[u] | \Psi_u \rangle + \langle \Psi_u | \sum_i (v-u)(\mathbf{r}_i) | \Psi_u \rangle && \text{same } \hat{T}, \hat{W} \\
&= E_u + \int [v(\mathbf{r}) - u(\mathbf{r})] n_0(\mathbf{r}) \, \mathrm{d}^3 r && \text{same density.}
\end{aligned}
$$

Key result

The many-body matrix element of the potential difference is exactly a one-density integral. No determinant factorization or mean-field approximation has been used.

HK I: reverse the roles, add, and reach a contradiction

The two inequalities are

$$E_v < E_u + \int (v-u) n_0 \, \mathrm{d}^3 r, \qquad E_u < E_v + \int (u-v) n_0 \, \mathrm{d}^3 r.$$

Adding cancels the density integrals:

$$E_v + E_u < E_u + E_v,$$

which is impossible. Thus

$$\boxed{n_0 \;\longmapsto\; v \pmod{C} \;\longmapsto\; \hat{H}[v] \pmod{NC}.}$$

With a nondegenerate ground state, the Hamiltonian fixes Ψ_0 up to a phase, and therefore all its expectation values.

Many-body connection

This is uniqueness in principle. It neither reconstructs v from measured n nor supplies an efficient way to solve for Ψ_0.

Degeneracy: replace strictness by an equality argument

For two ground states with the same density, define

$$A = \langle \hat{H}[v] \rangle_{\Psi_u} - E_v \geq 0, \qquad B = \langle \hat{H}[u] \rangle_{\Psi_v} - E_u \geq 0.$$

The equal-density identity gives

$$A + B = \int (v - u) n_0 \, \mathrm{d}^3 r + \int (u - v) n_0 \, \mathrm{d}^3 r = 0.$$

Hence $A = B = 0$: each trial state belongs to both ground-state eigenspaces. The common-state lemma again implies $v - u = C$.

- Potential uniqueness survives under the same continuum assumptions.
- The density need not select one pure state within a degenerate subspace.
- For ensembles use traces; equality means support only in the ground eigenspace. Constrained search below handles the variational problem directly.

Which densities may we vary? Three different questions

N-representable: some normalized antisymmetric N-particle state with finite kinetic energy produces n.

Interacting v-representable: a ground state of $\hat{T} + \hat{W} + \hat{V}[v]$ for an admissible local v produces n.

Noninteracting v-representable: a ground state of $\hat{T} + \hat{V}[v_s]$ for a common local v_s produces n.

A standard finite-kinetic-energy density domain is

$$\mathcal{I}_N = \{n \geq 0 : \int n = N,\ \sqrt{n} \in H^1(\mathbb{R}^3)\}.$$

H^1 requires square integrability of $\sqrt{n}$ and its weak gradient.

A trap to avoid

"Every positive function integrating to N" is too broad. N-representability does not imply either form of local-potential ground-state representability. HK does not prove noninteracting v-representability.

Levy constrained search: define the missing internal energy

Partition all wave functions according to their density:

$$\boxed{F_{\mathrm{LL}}[n] = \inf_{\Psi \to n} \langle \Psi | \hat{T} + \hat{W} | \Psi \rangle.}$$

- Search over normalized antisymmetric finite-energy states giving the specified n; assign $+\infty$ outside the admissible domain.
- Many different states may have that density. Select the least internal energy, rather than claiming they all have the same energy.
- No external potential appears: F_{LL} is universal for fixed statistics, kinetic operator, and interaction.
- We write an infimum safely; under the standard Coulomb-domain conditions the constrained minimum is attained.

Many-body connection

The many-body problem remains inside the constrained search. This extends the domain of the functional without requiring an explicit inverse map from a trial density to a potential.

HK II: regroup Rayleigh–Ritz into a density minimization

Start with the exact wave-function search:

$$
\begin{aligned}
E_0[v] &= \inf_{\Psi} \left\{ \langle \hat{T} + \hat{W} \rangle_\Psi + \int v n_\Psi \, \mathrm{d}^3 r \right\} \\
&= \inf_{n \in \mathcal{I}_N} \inf_{\Psi \to n} \left\{ \langle \hat{T} + \hat{W} \rangle_\Psi + \int v n \, \mathrm{d}^3 r \right\} \\
&= \boxed{ \inf_{n \in \mathcal{I}_N} \left\{ F_{\mathrm{LL}}[n] + \int v(\mathbf{r}) n(\mathbf{r}) \, \mathrm{d}^3 r \right\} . }
\end{aligned}
$$

The external term is constant within each class of states sharing n and can be taken outside the inner search.

Key result

The equality follows by partitioning the same set of trial states. It does not rely on finding a potential for every trial density.

HK II: prove the lower bound and prove it is attained

For every admissible n and every $\Psi \to n$,

$$\langle \hat{T} + \hat{W} \rangle_\Psi + \int v n \, \mathrm{d}^3 r \geq E_0[v].$$

Take the constrained infimum:

$$E_v[n] \equiv F_{\mathrm{LL}}[n] + \int v n \, \mathrm{d}^3 r \geq E_0[v].$$

Now insert the actual ground state $\Psi_0 \to n_0$:

$$F_{\mathrm{LL}}[n_0] \leq \langle \hat{T} + \hat{W} \rangle_{\Psi_0}, \qquad E_v[n_0] \leq E_0[v].$$

Combining the bounds yields

$$\boxed{E_v[n_0] = E_0[v] = \min_n E_v[n].}$$

A unique ground state gives a unique minimizing density. Degenerate ground states can give several minimizing densities.

Ensembles and the convex functional: keep the definitions distinct

An N-particle ensemble is $\hat{\mathcal{D}} \geq 0$, $\mathrm{Tr}\,\hat{\mathcal{D}} = 1$. Define

$$F_{\mathrm{ens}}[n] = \inf_{\hat{\mathcal{D}} \to n} \mathrm{Tr}[\hat{\mathcal{D}}(\hat{T} + \hat{W})].$$

Mixing ensembles mixes their densities and energies, so

$$F_{\mathrm{ens}}[\theta n_1 + (1-\theta) n_2] \leq \theta F_{\mathrm{ens}}[n_1] + (1-\theta) F_{\mathrm{ens}}[n_2].$$

In the standard Coulomb framework, Lieb's convex formulation is

$$F_L[n] = \sup_v \left\{ E_0[v] - \int v n \, \mathrm{d}^3 r \right\} = F_{\mathrm{ens}}[n] \leq F_{\mathrm{LL}}[n].$$

They agree at pure-state ground-state densities, but need not agree everywhere.

Key result

Below, F denotes the ensemble constrained-search functional. It gives the same physical ground-state energy and accommodates degenerate occupations. Do not silently identify a mixed state with one wave function.

The Euler equation: what the functional derivative means

Assume the needed derivatives and admissible variations exist. At a minimum, $\int \delta n \, \mathrm{d}^3 r = 0$ gives

$$0 = \delta E_v = \int \left[\frac{\delta F}{\delta n(\mathbf{r})} + v(\mathbf{r}) \right] \delta n(\mathbf{r}) \, \mathrm{d}^3 r.$$

For a sufficient class of two-sided admissible variations, the bracket is constant:

$$\boxed{\frac{\delta F}{\delta n(\mathbf{r})} + v(\mathbf{r}) = \mu.}$$

Equivalently vary $E_v[n] - \mu(\int n - N)$. On a fixed-N domain the derivative is defined only up to a constant, reflected in μ.

A trap to avoid

Existence of a minimizer does not guarantee differentiability. The convex statement is $-v \in \partial F[n_0]$, modulo the fixed-N constant: $F[n] \geq F[n_0] - \int v(n - n_0)$. At integer particle numbers, one-sided chemical potentials may differ.

What has been established, and what remains unsolved

Exact statements

- A ground-state density determines the scalar potential modulo a constant.
- Constrained search defines a universal internal-energy functional.
- Minimizing the exact functional yields the exact energy and a ground-state density.

Not supplied by these proofs

- An explicit or inexpensive formula for $F[n]$.
- A local noninteracting potential for every admissible density.
- A practical direct calculation of excitation energies.

Knowing the Hamiltonian fixes its spectrum in principle. The ground-state variational procedure does not itself compute that spectrum; auxiliary KS eigenvalues need separate interpretation.

Pause to think: locate the assumptions

Discuss with a neighbor

1. Why does the contradiction not distinguish v from $v + C$?
2. Does a degenerate ground-state density specify a unique wave function?
3. Why is minimizing over all normalized positive functions not a complete statement of the density variational principle?
4. Does constrained search provide an explicit formula for $F[n]$?

Answer: uniqueness, domain, and computation are different issues

1. The states are unchanged by NC; in the contradiction the trial state then need not give a strict inequality.
2. No. Potential uniqueness can fix the Hamiltonian and its ground-state space without selecting a particular state in that space.
3. Trial densities must come from admissible finite-energy states; otherwise the constrained-search energy is not finite.
4. No. It defines the exact quantity and proves its variational role. Evaluating the constrained search is still a many-body calculation.

Key result

Kohn–Sham addresses the computational representation. Approximating the exchange-correlation functional is a later, separate step.

Why put orbitals back into a density theory?

The local Thomas–Fermi kinetic approximation in 3D is

$$T_{\mathrm{TF}}[n] = \frac{3}{10}(3\pi^2)^{2/3} \int n^{5/3}\, \mathrm{d}^3 r$$

for an unpolarized electron gas. It has no explicit orbital shell structure. Kohn and Sham instead compute an auxiliary kinetic energy from orbitals:

$$T_s = \sum_i f_i \langle \phi_i | -\frac{1}{2}\nabla^2 | \phi_i \rangle, \qquad n(\mathbf{r}) = \sum_i f_i \sum_\sigma |\phi_i(\mathbf{r}\sigma)|^2.$$

- T_s is exact for the auxiliary noninteracting problem, not generally equal to the interacting kinetic energy T.
- Orbitals represent T_s implicitly as a functional of n.
- They recover noninteracting shell structure while retaining the density as the target quantity.

Define the noninteracting kinetic functional carefully

Using orthonormal spin orbitals and ensemble occupations,

$$\boxed{T_s[n] = \inf_{\{f_i,\phi_i\}\to n} \sum_i f_i \langle \phi_i | \hat{t} | \phi_i \rangle,} \qquad 0 \le f_i \le 1, \quad \sum_i f_i = N.$$

Restricting this search to integer occupations gives the determinant constrained search with density n.

Representation versus ground-state representation

Writing a density using orbitals is weaker than finding a common local potential whose noninteracting ground state has that density. The differential KS construction assumes a suitable supporting potential and derivatives, or their ensemble/subgradient counterparts.

We first derive the orbital equations for unit occupations of the occupied spin orbitals. Degenerate and spin-resolved occupations are added afterward.

Define exchange-correlation so the energy decomposition is exact

For a fixed symmetric interaction w, set

$$E_H[n] = \frac{1}{2}\iint n(\mathbf{r})w(\mathbf{r},\mathbf{r}')n(\mathbf{r}')\,\mathrm{d}^3r\,\mathrm{d}^3r', \qquad \boxed{E_{\mathrm{xc}}[n] = F[n] - T_s[n] - E_H[n].}$$

Therefore

$$\boxed{E_v[n] = T_s[n] + \int v_{\mathrm{ext}}\, n\,\mathrm{d}^3r + E_H[n] + E_{\mathrm{xc}}[n].}$$

At a constrained-search minimizing interacting state,

$$E_{\mathrm{xc}} = (T - T_s) + (W - E_H).$$

- Correlation includes a kinetic contribution. It is not just the pair-density cumulant.
- No approximation has yet been made, and no general theorem says this remainder is small.
- In practice replace E_{xc} by a chosen approximation and solve the resulting problem consistently.

Derive the Hartree potential: vary both density factors

For symmetric $w(\mathbf{r},\mathbf{r}')$,

$$\begin{aligned}\delta E_H &= \frac{1}{2}\iint w(\mathbf{r},\mathbf{r}')[\delta n(\mathbf{r})n(\mathbf{r}') + n(\mathbf{r})\delta n(\mathbf{r}')]\,\mathrm{d}^3r\,\mathrm{d}^3r' \\ &= \int\left[\int w(\mathbf{r},\mathbf{r}')n(\mathbf{r}')\,\mathrm{d}^3r'\right]\delta n(\mathbf{r})\,\mathrm{d}^3r.\end{aligned}$$

Thus

$$\boxed{v_H(\mathbf{r}) = \frac{\delta E_H}{\delta n(\mathbf{r})} = \int w(\mathbf{r},\mathbf{r}')n(\mathbf{r}')\,\mathrm{d}^3r',} \qquad \int n v_H = 2E_H.$$

For 3D Coulomb electrons, $w = 1/|\mathbf{r}-\mathbf{r}'|$. For the softened 1D model later, use its own interaction kernel; the formal variation is unchanged.

The variational object includes all orthonormality constraints

For occupied unit-occupation spin orbitals,

$$n(\mathbf{r}) = \sum_{i\in\mathrm{occ}} \sum_{\sigma} |\phi_i(\mathbf{r}\sigma)|^2, \qquad \langle\phi_i|\phi_j\rangle = \delta_{ij}.$$

Write the orbital energy and constrained Lagrangian:

$$\mathcal{E}[\{\phi_i\}] = \sum_{i\in\mathrm{occ}} \langle\phi_i|\hat{t}|\phi_i\rangle + E_{\mathrm{ext}}[n] + E_H[n] + E_{\mathrm{xc}}[n],$$

$$\mathcal{L} = \mathcal{E} - \sum_{ij} \Lambda_{ij}(\langle\phi_j|\phi_i\rangle - \delta_{ji}), \qquad \Lambda = \Lambda^\dagger.$$

Many-body connection

Use a matrix of multipliers, not independent scalar orbital energies at the outset. Off-diagonal constraints enforce mutual orthogonality, just as in the Hartree–Fock variation.

Treat ϕ_i and ϕ_i^* as independent variables during variation.

Carry out each orbital derivative

For the variation with respect to $\phi_i^*(x)$,

$$\frac{\delta n(\mathbf{r})}{\delta \phi_i^*(x)} = \delta^{(3)}(\mathbf{r} - \mathbf{r}_x)\phi_i(x), \qquad x = (\mathbf{r}_x, \sigma_x).$$

The contributions are

$$\begin{aligned}
\frac{\delta T_s}{\delta \phi_i^*(x)} &= \hat{t}\phi_i(x), \\
\frac{\delta E_{\text{ext}}}{\delta \phi_i^*(x)} &= v_{\text{ext}}(\mathbf{r})\phi_i(x), \\
\frac{\delta (E_H + E_{\text{xc}})}{\delta \phi_i^*(x)} &= [v_H(\mathbf{r}) + v_{\text{xc}}(\mathbf{r})]\phi_i(x), \qquad v_{\text{xc}} = \frac{\delta E_{\text{xc}}}{\delta n}, \\
\frac{\delta}{\delta \phi_i^*(x)} \sum_{jk} \Lambda_{jk} \langle \phi_k | \phi_j \rangle &= \sum_j \Lambda_{ji}\phi_j(x).
\end{aligned}$$

Here a density-functional derivative and the chain rule turn the interaction functional into an orbital equation.

Stationarity and the canonical Kohn–Sham equations

Setting the derivative of $\mathcal{L}$ to zero gives

$$\hat{h}_s\phi_i = \sum_j \Lambda_{ji}\phi_j, \qquad \hat{h}_s = -\frac{1}{2}\nabla^2 + v_s(\mathbf{r}),$$

$$v_s = v_{\text{ext}} + v_H[n] + v_{\text{xc}}[n].$$

The density and energy are invariant under unitary rotations among equally occupied orbitals. Diagonalize the Hermitian Λ in that subspace:

$$\boxed{\left[-\frac{1}{2}\nabla^2 + v_{\text{ext}} + v_H[n] + v_{\text{xc}}[n]\right]\phi_i = \epsilon_i\phi_i, \qquad n = \sum_{i\in\text{occ},\sigma} |\phi_i|^2.}$$

Key result

The problem is nonlinear: the density defines the potential, the potential defines the orbitals, and the occupied orbitals define the density.

Check the construction against the density Euler equation

For the auxiliary noninteracting ground state,

$$\frac{\delta T_s}{\delta n} + v_s = \mu_s.$$

For the interacting density minimization,

$$\frac{\delta F}{\delta n} + v_{\mathrm{ext}} = \mu.$$

Since $F = T_s + E_H + E_{\mathrm{xc}}$, subtraction gives

$$v_s(\mathbf{r}) = v_{\mathrm{ext}}(\mathbf{r}) + v_H(\mathbf{r}) + v_{\mathrm{xc}}(\mathbf{r}) + \mu_s - \mu.$$

The final constant is absorbed into the potential gauge.

Two ways to see the same construction

The constrained orbital variation yields the differential equations. The density Euler equations explain why the effective potential has exactly these three contributions, when the required derivatives exist.

Occupations and spin: avoid hidden factors of two

For spin orbitals, $0 \le f_i \le 1$ and $\sum_i f_i = N$. At zero temperature, occupation minimization fills the lowest levels:

$$f_i = 1 \ (\epsilon_i < \mu), \quad f_i = 0 \ (\epsilon_i > \mu), \quad 0 \le f_i \le 1 \ (\epsilon_i = \mu).$$

For collinear spin DFT,

$$n_\sigma(\mathbf{r}) = \sum_i f_{i\sigma} |\phi_{i\sigma}(\mathbf{r})|^2, \quad v_{\mathrm{xc},\sigma} = \frac{\delta E_{\mathrm{xc}}[n_\uparrow, n_\downarrow]}{\delta n_\sigma}, \quad n = n_\uparrow + n_\downarrow.$$

- Doubly occupied *spatial* orbitals use occupations up to 2 instead. Do not also sum spin a second time.
- The numerical model later is *spinless*: one particle per orbital.
- Finite-temperature Fermi occupations minimize a free energy containing an entropy term. Smearing is not an undocumented zero-temperature identity.

Derive the total energy: why the eigenvalue sum is wrong

Multiply each KS equation by $f_i \phi_i^*$, integrate, and sum:

$$\sum_i f_i \epsilon_i = T_s + \int n v_{\mathrm{ext}} + \int n v_H + \int n v_{\mathrm{xc}} = T_s + E_{\mathrm{ext}} + 2E_H + \int n v_{\mathrm{xc}}.$$

Subtract the actual energy $E = T_s + E_{\mathrm{ext}} + E_H + E_{\mathrm{xc}}$:

$$\boxed{E = \sum_i f_i \epsilon_i - E_H[n] + E_{\mathrm{xc}}[n] - \int n(\mathbf{r}) v_{\mathrm{xc}}(\mathbf{r}) \, \mathrm{d}^3 r.}$$

For spin DFT replace the last integral by $\sum_\sigma \int n_\sigma v_{\mathrm{xc},\sigma}$.

- The Hartree potential counts each pair twice in the eigenvalue sum.
- v_{xc} is a derivative; its expectation is not E_{xc}.
- Add E_{NN} for clamped-nuclei molecular/crystal total energies.

This is a consistency identity for the same orbitals, potential, and density.

Local operator does not mean a local density functional

In conventional KS, $v_{\mathrm{xc}}([n];\mathbf{r})$ multiplies an orbital at $\mathbf{r}$:

$$(\hat{v}_{\mathrm{xc}}\phi)(\mathbf{r}) = v_{\mathrm{xc}}([n];\mathbf{r})\phi(\mathbf{r}).$$

Its value may nevertheless depend on the density everywhere.

Object	Dependence	Action on an orbital
$v_H[n](\mathbf{r})$	integral over $n(\mathbf{r}')$	multiplicative
$v_{\mathrm{xc}}^{\mathrm{LDA}}$	value of n at $\mathbf{r}$	multiplicative
$v_{\mathrm{xc}}^{\mathrm{GGA}}$	n and spatial derivatives	multiplicative
Exact local KS v_{xc}	generally nonlocal in n	multiplicative
HF Fock exchange	off-diagonal one-body RDM	integral operator

A trap to avoid

Exact KS is not inherently approximate. LDA/GGA are approximation choices for its unknown functional. Nonlocal density dependence and a nonlocal orbital operator are different meanings of "nonlocal".

LDA: differentiate the energy per volume, not per particle

Let $\epsilon_{\mathrm{xc}}(n)$ be the uniform-gas energy *per particle*. The LDA approximation is

$$E_{\mathrm{xc}}^{\mathrm{LDA}}[n] = \int f_{\mathrm{xc}}(n(\mathbf{r}))\,\mathrm{d}^3 r, \qquad f_{\mathrm{xc}}(n) = n\epsilon_{\mathrm{xc}}(n).$$

Varying at each point gives

$$\delta E_{\mathrm{xc}} = \int [\epsilon_{\mathrm{xc}}(n) + n\epsilon'_{\mathrm{xc}}(n)]\delta n\,\mathrm{d}^3 r,$$

$$\boxed{v_{\mathrm{xc}}^{\mathrm{LDA}}(\mathbf{r})} \boxed{= \frac{\mathrm{d}[n\epsilon_{\mathrm{xc}}(n)]}{\mathrm{d}n} = \epsilon_{\mathrm{xc}}(n) + n\epsilon'_{\mathrm{xc}}(n).}$$

If correlation is parameterized by $r_s \propto n^{-1/3}$ in 3D,

$$v_c = \epsilon_c(r_s) - \frac{r_s}{3}\frac{\mathrm{d}\epsilon_c}{\mathrm{d}r_s}.$$

Energy and potential must be derived from the *same* functional or interpolant.

Derive the 3D exchange potential and check the units

For a spin-unpolarized 3D gas,

$$k_F = (3\pi^2 n)^{1/3}, \qquad \epsilon_x = -\frac{3k_F}{4\pi} = -\frac{3}{4}\left(\frac{3}{\pi}\right)^{1/3} n^{1/3}.$$

Therefore

$$E_x^{\mathrm{LDA}} = -C_x \int n^{4/3}\,\mathrm{d}^3 r, \quad C_x = \frac{3}{4}\left(\frac{3}{\pi}\right)^{1/3}, \quad \boxed{v_x = -\left(\frac{3}{\pi}\right)^{1/3} n^{1/3} = \frac{4}{3}\epsilon_x.}$$

With $n = 3/(4\pi r_s^3)$,

$$\epsilon_x = -\frac{0.4581653}{r_s}\ \mathrm{Ha} = -\frac{0.9163306}{r_s}\ \mathrm{Ry}.$$

A trap to avoid

This is the 3D, two-spin-species formula. The 1D spinless model requires a different uniform reference and functional, not a reused $n^{1/3}$ law.

GGA: show the integration by parts explicitly

For $E_{\mathrm{xc}}^{\mathrm{GGA}} = \int f(n, \nabla n)\, \mathrm{d}^3 r$,

$$\begin{aligned}\delta E_{\mathrm{xc}} &= \int \left[\frac{\partial f}{\partial n} \delta n + \frac{\partial f}{\partial (\nabla n)} \cdot \nabla \delta n \right] \mathrm{d}^3 r \\ &= \int \left[\frac{\partial f}{\partial n} - \nabla \cdot \frac{\partial f}{\partial (\nabla n)} \right] \delta n \, \mathrm{d}^3 r + \text{surface term.}\end{aligned}$$

For appropriate decaying or periodic boundary conditions,

$$\boxed{v_{\mathrm{xc}}^{\mathrm{GGA}} = \frac{\partial f}{\partial n} - \nabla \cdot \frac{\partial f}{\partial (\nabla n)}.}$$

For $q = |\nabla n|^2$ and $f = f(n, q)$, this becomes $v_{\mathrm{xc}} = f_n - 2\nabla \cdot (f_q \nabla n)$.

Many-body connection

The boundary condition and discrete derivative must match the energy discretization. A continuum derivative copied into an inconsistent numerical scheme can destroy variational consistency.

Hartree–Fock, exact-exchange KS, and generalized KS

- **HF:** minimize the expectation of the true Hamiltonian over determinants. The exchange variation gives a nonlocal Fock operator.
- **Conventional KS:** reproduce the density using a multiplicative potential. Exact E_{xc} would be exact within the representation assumptions.
- **Exact exchange via OEP:** an orbital exchange functional can be used with a local optimized effective potential; it need not yield HF orbitals.
- **Hybrids/generalized KS:** direct orbital variation of an exchange mixture usually produces a nonlocal operator. Do not apply local-KS energy or spectral formulas unchanged.

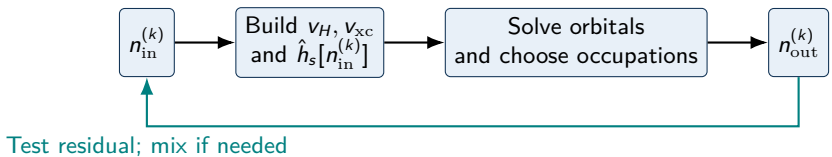
A trap to avoid

There is no general theorem that the same fixed percentage of exchange is recovered, that local-exchange energy always lies above HF, or that any hybrid completely removes self-interaction. Such comparisons depend on the functional, system, and representation.

The nonlinear problem is a fixed point

Define the output-density map $\mathcal{F}[n]$ by building and solving the KS Hamiltonian at an input density. The target satisfies

$$\boxed{n_* = \mathcal{F}[n_*].}$$



At fixed occupations, the solve is linear; the repeated density–potential update is the nonlinear part. A small eigenproblem residual alone is not self-consistency.

Basis implementation: the generalized eigenvalue problem

Expand $\phi_i(x) = \sum_\mu C_{\mu i}\chi_\mu(x)$. Define

$$S_{\mu\nu} = \int \chi_\mu^* \chi_\nu \, \mathrm{d}x, \quad H_{\mu\nu}[n] = \int \chi_\mu^* (\hat{t} + v_{\mathrm{ext}} + v_H + v_{\mathrm{xc}})\chi_\nu \, \mathrm{d}x.$$

The projected equations and normalization are

$$\boxed{HC = SC\varepsilon, \qquad C^\dagger SC = I.}$$

With $D_{\mu\nu} = \sum_i f_i C_{\mu i} C_{\nu i}^*$,

$$n(\mathbf{r}) = \sum_{\mu\nu,\sigma} \chi_\mu(\mathbf{r}\sigma) D_{\mu\nu} \chi_\nu^*(\mathbf{r}\sigma), \qquad \mathrm{Tr}(DS) = N.$$

For $S > 0$, orthogonalize with $X = S^{-1/2}$ and solve $X^\dagger H X$. Remove or control nearly dependent basis directions before taking $S^{-1/2}$.

Grid implementation: quadrature weights belong in the density

For a uniform 1D grid x_j with spacing h and Dirichlet boundaries,

$$-\frac{1}{2}\phi''(x_j) \simeq \frac{2\phi_j - \phi_{j-1} - \phi_{j+1}}{2h^2}.$$

If the matrix eigensolver normalizes $\sum_j |u_{ji}|^2 = 1$, then

$$\phi_i(x_j) = u_{ji}/\sqrt{h}, \qquad n_j = \frac{1}{h}\sum_i f_i |u_{ji}|^2, \qquad h\sum_j n_j = N.$$

For general quadrature weights w_j, use the corresponding weighted overlap or transformed coefficients consistently.

A trap to avoid

Do not normalize orbitals with one quadrature and integrate density or energy with another without checking consistency. Refine the grid and enlarge the box independently; a converged SCF loop cannot repair either error.

Build the Hartree field with the correct kernel and boundaries

In a quadrature representation,

$$(v_H)_j = \sum_k w_k \, w(x_j, x_k) n_k, \qquad E_H = \frac{1}{2} \sum_j w_j n_j (v_H)_j.$$

For Coulomb electrons in *three dimensions*, an equivalent formulation is

$$\nabla^2 v_H(\mathbf{r}) = -4\pi n(\mathbf{r}), \qquad v_H(\mathbf{r}) \sim N/|\mathbf{r}| \quad \text{for an isolated localized density.}$$

- Isolated systems need isolated/open-boundary electrostatics.
- Periodic cells require a consistent neutrality and zero-mode convention. Include the ion/background terms in a physical periodic total energy.
- The softened 1D kernel $1/\sqrt{(x-x')^2 + a^2}$ is a convolution; it is not the solution of this 3D Poisson equation on a line.

Density mixing: stability follows from a linearized map

Let the unmixed residual be $R^{(k)} = n_{\mathrm{out}}^{(k)} - n_{\mathrm{in}}^{(k)}$. Linear mixing uses

$$n_{\mathrm{in}}^{(k+1)} = n_{\mathrm{in}}^{(k)} + \alpha R^{(k)} = (1-\alpha) n_{\mathrm{in}}^{(k)} + \alpha n_{\mathrm{out}}^{(k)}.$$

Near a fixed point, with $J = \delta\mathcal{F}/\delta n$,

$$\delta n^{(k+1)} \simeq [(1-\alpha) I + \alpha J] \delta n^{(k)}.$$

Local linear convergence is guaranteed when the spectral radius is less than one on allowed variations.

- $0 < \alpha \leq 1$ preserves positivity and particle number if both inputs have them.
- Small α can damp oscillations; it is not a universal convergence guarantee.
- Pulay/DIIS and preconditioned mixing accelerate difficult cases but need safeguards.
- Test R, not just the mixed step αR, which can be made small artificially.

A practical algorithm, with consistent bookkeeping

```
n_in = initial_density(N) # nonnegative; integral = N
E_previous = None
for k in range(max_iterations):
    vH, vxc = potentials(n_in)
    eps, orbitals = solve_KS(vext + vH + vxc)
    f = occupations(eps, N) # use the stated spin convention
    n_out = density(orbitals, f)
    R = n_out - n_in
    E = direct_energy(orbitals, f, n_out)
    if residual_small(R) and energy_stable(E, E_previous):
        break
    n_in = (1-alpha)*n_in + alpha*n_out
    E_previous = E
else:
    raise ConvergenceError
```

Rebuild final potentials and check orbital/energy consistency. A maximum-iteration exit is a failure report, not convergence.

During SCF, compute an energy from one consistent output state

Orbitals obtained from $\hat{h}_s[n_{\mathrm{in}}]$ give n_{out}. Evaluate the direct functional as

$$E^{(k)} = \sum_i f_i \langle \phi_i^{(k)} | \hat{t} + v_{\mathrm{ext}} | \phi_i^{(k)} \rangle + E_H[n_{\mathrm{out}}^{(k)}] + E_{\mathrm{xc}}[n_{\mathrm{out}}^{(k)}].$$

This is the chosen functional evaluated on the actual output orbitals/density. Away from self-consistency,

$$\sum_i f_i \epsilon_i^{(k)} = T_s^{(k)} + E_{\mathrm{ext}}[n_{\mathrm{out}}] + \int n_{\mathrm{out}} (v_H[n_{\mathrm{in}}] + v_{\mathrm{xc}}[n_{\mathrm{in}}]).$$

A trap to avoid

The usual double-counting formula cannot combine old eigenvalues with newly mixed density corrections arbitrarily. At convergence, the direct and eigenvalue-corrected energies must agree within an energy-consistency tolerance compatible with the density and orbital residuals.

Convergence checks: three numerical questions, then physics

Check	Example diagnostic	What it tests
Particle number	$\lvert \int n - N \rvert$	occupations and quadrature
Orthonormality	$\lVert C^\dagger S C - I \rVert$	basis/eigensolver normalization
SCF consistency	$\lVert n_{\mathrm{out}} - n_{\mathrm{in}} \rVert$	nonlinear fixed point
Orbital stationarity	$\lVert HC - SC\varepsilon \rVert$	final KS equations
Energy bookkeeping	$\lvert E_{\mathrm{direct}} - E_{\mathrm{eig}} \rvert$	consistent functional terms
Representation	box/grid/basis refinement	discretization error

Also try different starting densities and relevant spin/symmetry sectors.

Key result

A self-consistent solution may be a local stationary solution, not the global functional minimum. Tiny numerical residuals do not establish the accuracy of the approximate exchange-correlation functional.

Use the Week-43 Hamiltonian, with its own exchange functional

Take N spinless fermions in oscillator units:

$$\hat{h}_0 = -\frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + \frac{x^2}{2}, \qquad w(s) = \frac{1}{\sqrt{s^2 + a^2}}, \qquad a = 0.5.$$

Use the first M harmonic-oscillator functions as a Galerkin basis. The numerical functional is

$$E = \sum_{i=1}^{N} \langle \phi_i | \hat{h}_0 | \phi_i \rangle + \frac{1}{2} \iint n(x) w(x-y) n(y)\, \mathrm{d}x\, \mathrm{d}y + \int e_x(n(x))\, \mathrm{d}x.$$

- Exchange-only LDA: no correlation functional is included.
- The uniform reference is 1D and spinless; each orbital holds one particle.
- Calculations below were recomputed for this supplement, using a refined uniform-exchange evaluation. Do not mix their last digits with older tables.

Derive the 1D uniform-gas exchange energy

The filled interval $-k_F \le k \le k_F$ gives

$$n = \int_{-k_F}^{k_F} \frac{\mathrm{d}k}{2\pi} = \frac{k_F}{\pi}, \qquad \gamma(s) = \int_{-k_F}^{k_F} \frac{\mathrm{d}k}{2\pi} e^{iks} = \frac{\sin(k_F s)}{\pi s}.$$

For a determinant, $P = n^2 - |\gamma(s)|^2$. Its exchange energy per length is

$$\begin{aligned} e_x(n) &= -\frac{1}{2}\int_{-\infty}^{\infty} w(s)|\gamma(s)|^2\,\mathrm{d}s \\ &= -\int_0^{\infty} \frac{1}{\sqrt{s^2+a^2}}\left[\frac{\sin(\pi n s)}{\pi s}\right]^2 \mathrm{d}s. \end{aligned}$$

$$\boxed{\epsilon_x(n) = e_x(n)/n, \qquad E_x^{\mathrm{LDA}} = \int e_x(n(x))\,\mathrm{d}x, \qquad v_x(n) = e_x'(n).}$$

The uniform Hartree energy needs a neutralizing background; the exchange integral is finite. The finite trapped Hartree term is evaluated directly.

Tabulate energy and derive the potential consistently

For $w(s) = 1/\sqrt{s^2 + a^2}$, its Fourier transform is $\widetilde{w}(q) = 2K_0(a|q|)$. With $Q = 2\pi n$,

$$e_x(n) = -\frac{1}{2\pi^2}\int_0^Q (Q - q)K_0(aq)\,\mathrm{d}q, \qquad v_x(n) = -\frac{1}{\pi}\int_0^Q K_0(aq)\,\mathrm{d}q.$$

K_0 is the modified Bessel function; its logarithmic endpoint singularity is integrable. The companion code tabulates e_x and its slope, then uses the derivative of the *same* Hermite interpolant in SCF.

n	$\epsilon_x(n)$	$v_x(n)$
0.05	−0.1735293	−0.2973727
0.20	−0.4233271	−0.6578909
0.80	−0.7588439	−0.9658763
2.50	−0.9189555	−0.9998969

Both tend to $-1/(2a) = -1$ at high density; neither follows the 3D power law.

Assemble the finite-basis matrices using one quadrature

On quadrature nodes x_j with weights w_j, let $B_{j\mu} = \chi_\mu(x_j)$. Use $S = B^\dagger W B$ and orthogonalize if needed. In the orthogonalized basis,

$$F[n] = h_0 + B^\dagger W \,\mathrm{diag}(v_H[n] + v_x[n]) B, \quad FC = C\varepsilon, \quad C^\dagger C = I.$$

For spinless integer occupations,

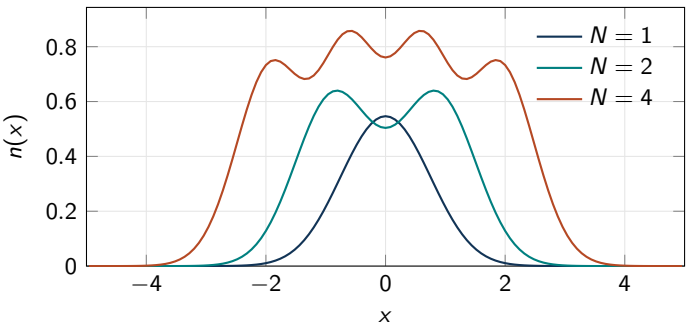
$$n_j = \sum_{i=1}^{N} |(BC)_{ji}|^2, \quad (v_H)_j = \sum_k w_k \frac{n_k}{\sqrt{(x_j - x_k)^2 + a^2}}.$$

The direct energy is

$$E = \mathrm{Tr}(D h_0) + \frac{1}{2} \sum_j w_j n_j (v_H)_j + \sum_j w_j e_x(n_j), \qquad D = C_{\mathrm{occ}} C_{\mathrm{occ}}^\dagger.$$

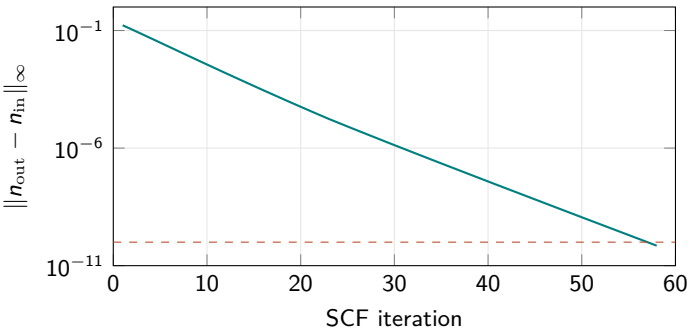
Baseline settings: $M = 8$, $x \in [-8, 8]$, 401 Simpson nodes, 601 exchange-table nodes up to $n = 3$, mixing $\alpha = 0.30$.

The computed densities retain orbital shell structure



- Fresh calculations with the stated exchange-only 1D functional.
- $N = 1, 2, 4$; $\int n = N$ to floating-point precision.
- The density is computed from occupied orbitals, rather than a local Thomas–Fermi kinetic-energy formula.
- Increasing N changes both occupations and the self-consistent field.

SCF convergence: use the unmixed fixed-point residual



- $N = 4$, $M = 8$, $\alpha = 0.30$.
- 58 iterations; final maximum density residual: 7.08×10^{-11}.
- Also require $|\Delta E| < 10^{-12}$ in oscillator energy units.
- A broadened initial density converges in 63 iterations to the same energy.

This is a check of this calculation, not a convergence guarantee for all systems.

Recomputed energies and checks for the eight-function example

N	$\langle \hat{h}_0 \rangle$	E_H	E_x^{LDA}	E
1	0.50063452	0.60855573	−0.56165318	0.54753706
2	2.02320483	1.97451757	−1.27748570	2.72023670
4	8.13378445	6.42090538	−2.86865934	11.68603049

For $N = 4$:

$$|\int n - 4| < 10^{-14}, \quad \|C_{\mathrm{occ}}^{\dagger} C_{\mathrm{occ}} - I\| < 5 \times 10^{-16}, \quad |E_{\mathrm{direct}} - E_{\mathrm{eig}}| = 1.09 \times 10^{-10}.$$

A density-space finite-difference check of $\delta(E_H + E_x)/\delta n$ agrees with $v_H + v_x$ to 6.9×10^{-11} for the tested direction.

A trap to avoid

These are energies of the stated approximate functional in a finite basis. They are not newly claimed exact interacting energies or independent measurements of functional accuracy.

Tiny SCF residual does not imply a converged representation

For $N = 4$, the same exchange-only functional gives

Change	Energy	Difference from baseline
$M = 8$, 401 nodes, $[-8, 8]$	11.6860304873	—
$M = 8$, 801 nodes, $[-8, 8]$	11.6860304873	2.5×10^{-12}
$M = 8$, 501 nodes, $[-10, 10]$	11.6860304873	1.8×10^{-13}
$M = 12$, 501 nodes, $[-10, 10]$	11.6853429697	-6.88×10^{-4}
$M = 16$, 501 nodes, $[-10, 10]$	11.6852955478	-7.35×10^{-4}

Doubling the exchange table from 601 to 1201 nodes changes the $M = 8$ energy by 2.2×10^{-11}.

Key result

Here grid and exchange-table errors are small compared with basis truncation. A still larger basis would be needed to establish a complete-basis limit; none of these checks eliminates the functional approximation.

One particle exposes self-interaction without a many-body ambiguity

For one particle, the physical two-body operator vanishes. In this trap, the exact ground-state energy is $E_0 = 1/2$. Exact exchange cancels Hartree for its one-particle determinant:

$$E_x = -E_H, \qquad E_c = 0.$$

The recomputed local-exchange result instead gives

$$E_H + E_x^{\mathrm{LDA}} = 0.04690255, \qquad \langle \hat{h}_0 \rangle - \frac{1}{2} = 0.00063452,$$

$$\boxed{E^{\mathrm{LDA}} - \frac{1}{2} = 0.04753706.}$$

- The first number is the self-interaction residue on the computed density.
- The second is orbital relaxation in h_0.
- For several particles, comparing local and determinant exchange on fixed orbitals does not cleanly separate locality from self-interaction.

What do Kohn–Sham orbitals and eigenvalues mean?

The auxiliary system targets the ground-state density; its determinant need not be the interacting ground-state wave function.

- KS eigenvalue differences are not generally neutral many-body excitation energies.
- For an isolated bound atom/molecule, the exact local functional with the proper potential zero gives $\epsilon_H = -I$ (approach integer N from below).
- The fundamental charge gap is

$$E_g = E(N+1) + E(N-1) - 2E(N) = \epsilon_L - \epsilon_H + \Delta_{\mathrm{xc}}.$$

- Δ_{xc} is the derivative discontinuity. Ordinary smooth LDA/GGA also have potential errors; not every gap error is solely the missing jump.

The isolated-system ionization statement is not a literal interpretation of the confining oscillator spectrum. Generalized-KS/hybrid spectra require their own analysis.

A reliable practical calculation separates four tasks

1. **Model:** specify particles, interaction, external potential, spin, dimensionality, boundaries, and any ionic energy.
2. **Functional:** state the approximation; derive potentials from that same energy. Check one-particle and uniform-reference limits.
3. **Numerics:** converge orbitals, the density fixed point, and the representation independently. Compare direct and eigenvalue-corrected energies.
4. **Physics:** examine competing spin/symmetry states and benchmark observables against reliable references where possible.

A trap to avoid

An approximate functional is variational with respect to itself, but generally supplies no rigorous upper bound to the exact interacting energy. A small residual is not an error bar for that functional.

Summary and the runnable companion

- **HK I:** equal ground-state densities cannot belong to inequivalent scalar potentials under the stated assumptions.
- **HK II:** constrained search proves $E_0 = \min_n [F[n] + \int vn]$.
- **KS:** define $E_{\mathrm{xc}} = F - T_s - E_H$, vary orthonormal orbitals, and obtain $v_s = v_{\mathrm{ext}} + v_H + v_{\mathrm{xc}}$.
- **Implementation:** build, solve, occupy, compare, mix, and verify; the total energy is not the raw eigenvalue sum.

```
python3 kohn_sham_1d_example.py --output-dir ks_example_results
```

The companion requires NumPy, SciPy, and Matplotlib. It reproduces the 1D tables, plots, functional-derivative check, and grid/basis tests in these slides. This LaTeX source embeds its plotted data and needs no external figures or original course folders to compile.

References: the foundations and mathematical qualifications

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References: approximations, eigenvalues, and course connection

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- L. J. Sham and M. Schlüter, *Density-Functional Theory of the Energy Gap*, Phys. Rev. Lett. **51**, 1888 (1983). doi:10.1103/PhysRevLett.51.1888.
- C. Glittum and M. Hjorth-Jensen, supplied `week42.tex` and `week43.tex`; M. Hjorth-Jensen, `BookPrograms/chapter08/dft.py`. These supply the visual template, course progression, and 1D teaching Hamiltonian.
- *From the many-body wave function to densities*, supplied one- and two-body RDM supplement. The ordered-pair convention and the distinction between the local density and the RDMs are retained here.